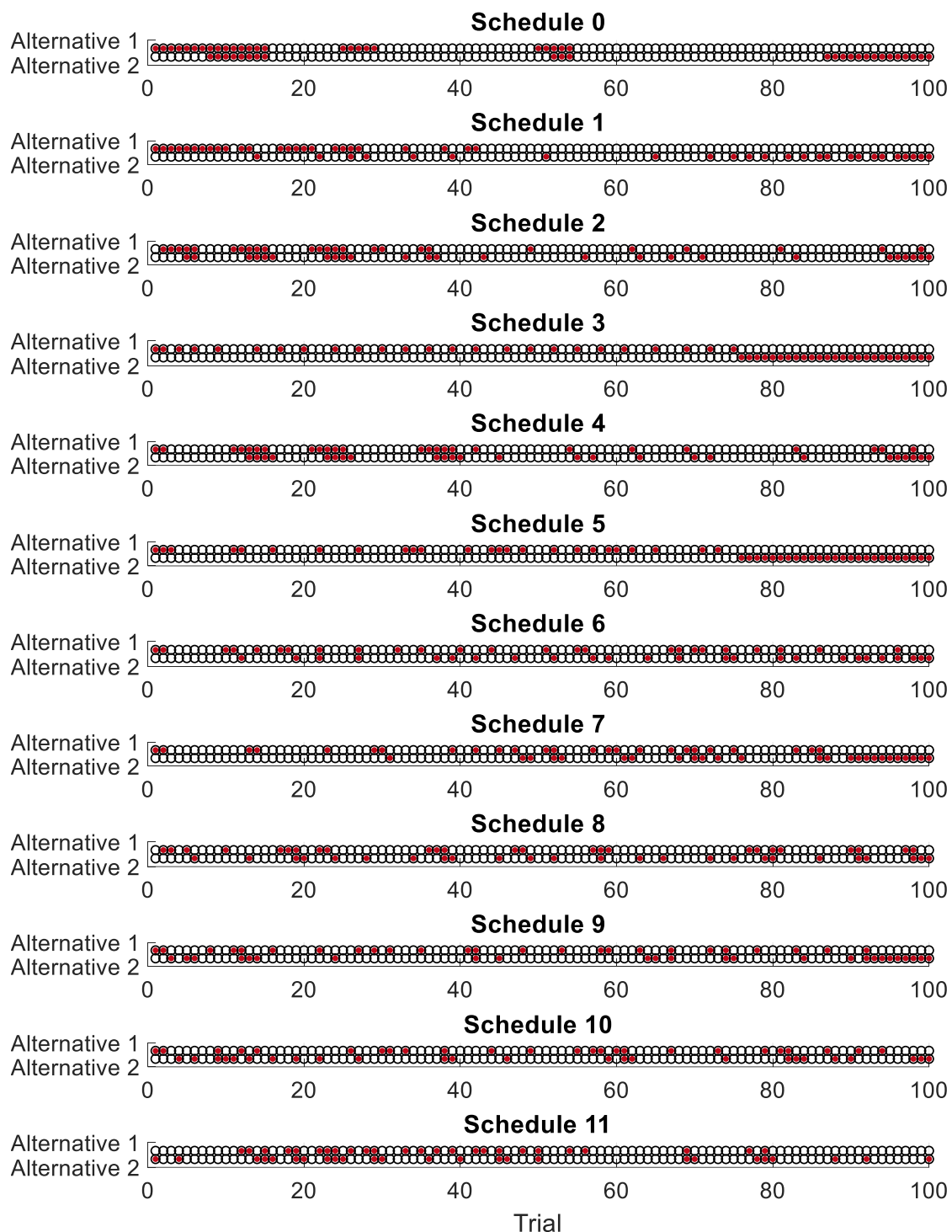


1 Behavior Engineering Using Quantitative Reinforcement Learning Models –

2 Supplementary Information



3

4 **Fig S1. The reward schedules that were tested on human participants.** Schedule

5 0 corresponds to the experimentally-optimized schedule. Other schedule indexes

6 correspond to Supplementary Data 1.

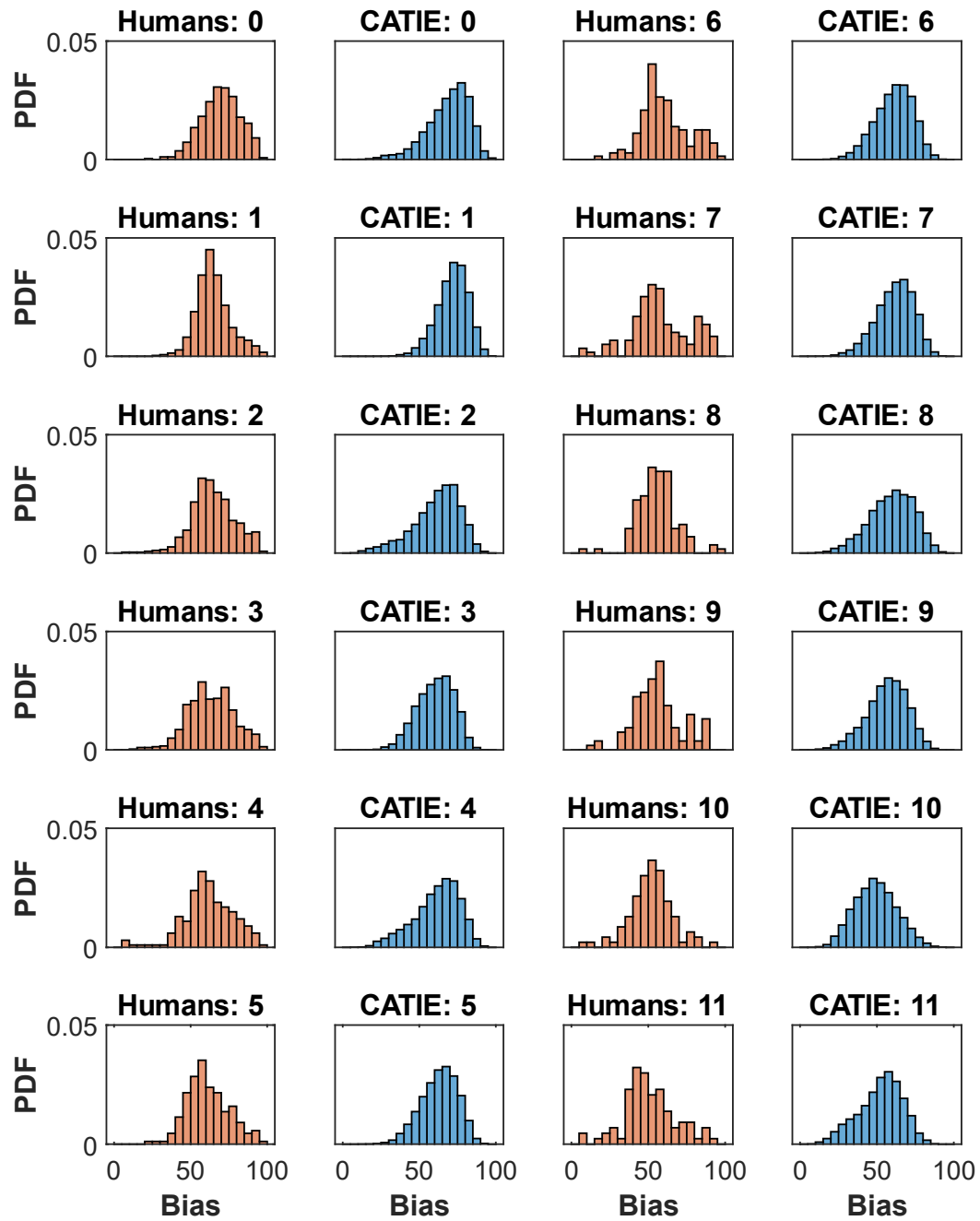
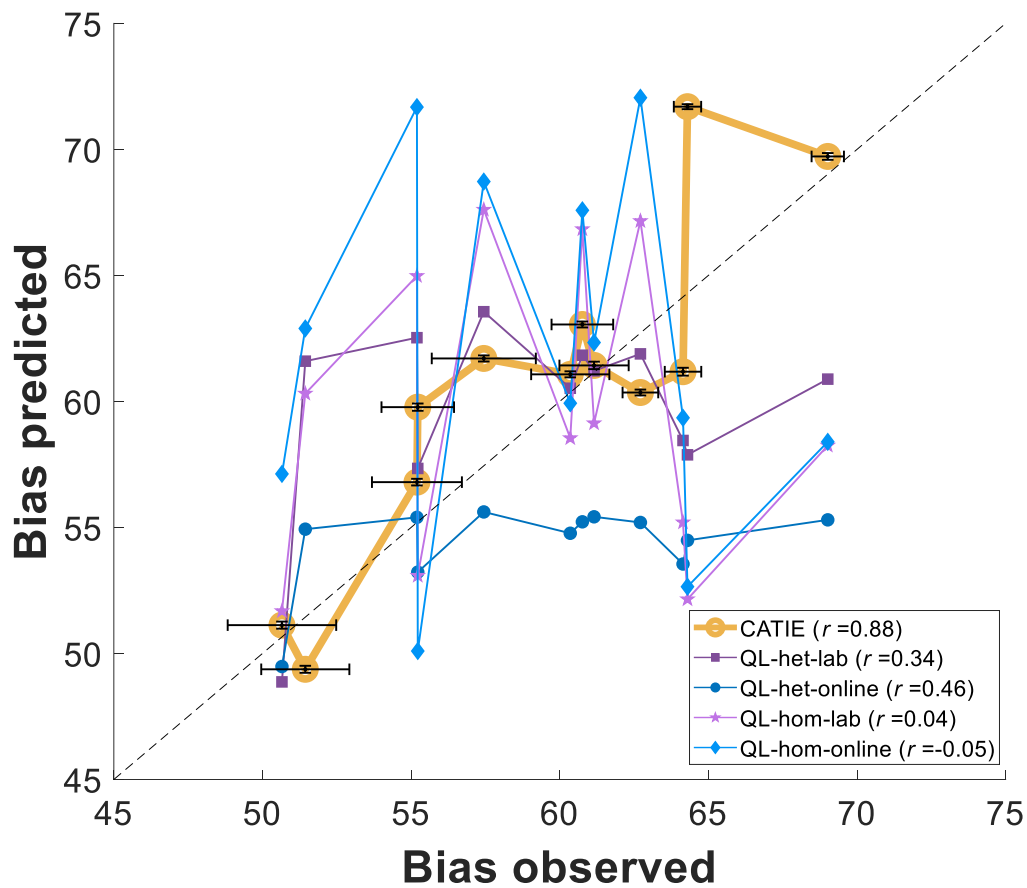
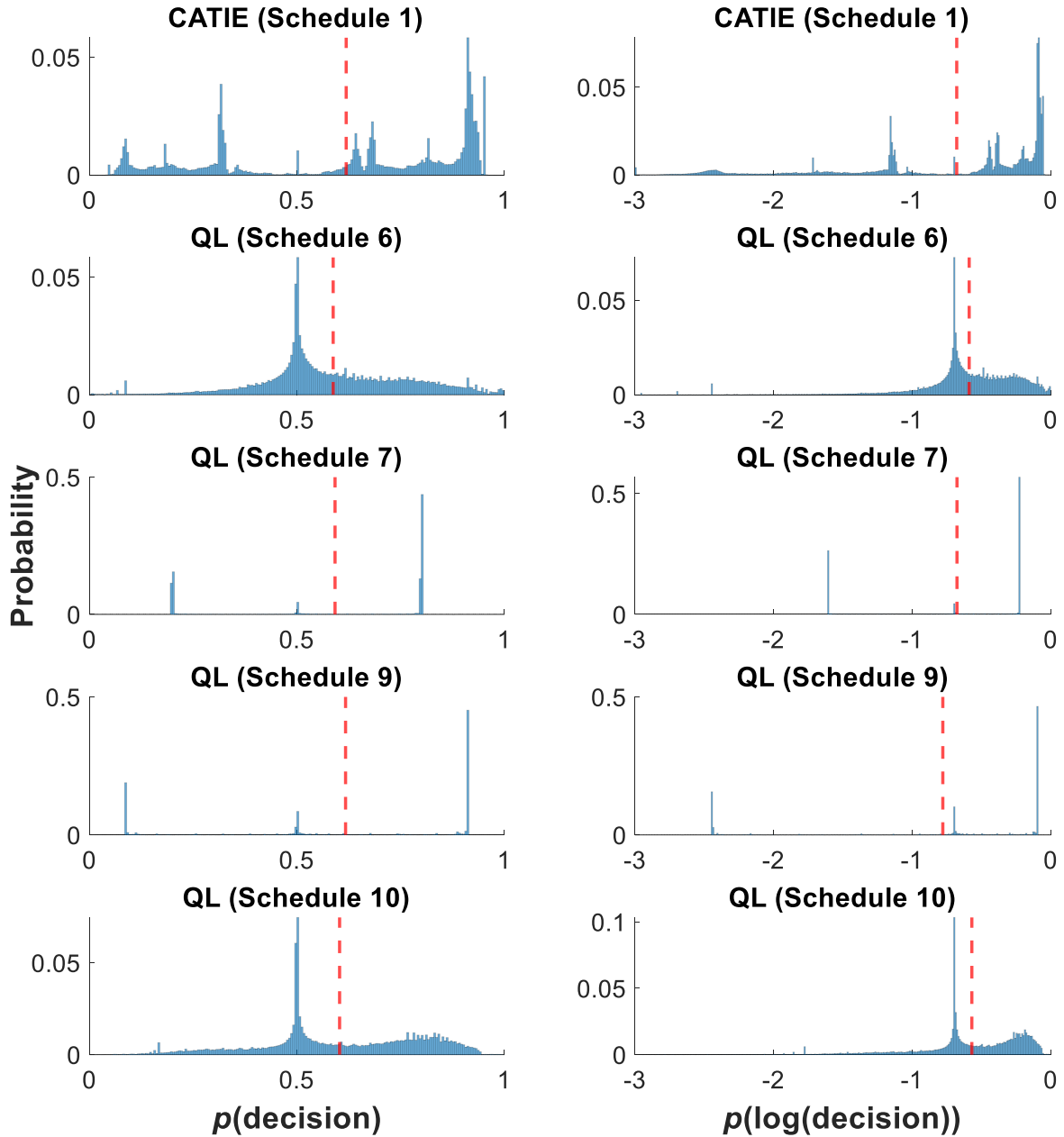


Fig S2 - Distributions of choice biases. Red, the distributions of biases of the human participants for each of the reward schedules in Fig. S1. The number of participants per distribution is different, and is specified in Supplementary Data 1. Blue, the distributions of the biases of 10,000 simulations of the CATIE model for the same reward schedules.



14

15 **Fig S3 - Correlation of behavioral bias with simulated bias.** Both axes present the
 16 average number of choices of the target side (the “bias”) for the 11 schedules tested
 17 in the competition, with the additional schedule 0 (see Results). The abscissa presents
 18 the sorted average bias as measured from behavior of the competition’s experimental
 19 participants (each data point is summarized over a different number of participants,
 20 see Supplementary Data 1). The ordinate presents the average bias for the same
 21 reward schedules as simulated 1,000 times by each of the 5 different quantitative
 22 models. The values of r in the legend depict the correlation coefficient of the predicted
 23 bias and empirical bias for each of the models. Error bars are standard error of the
 24 mean, centered around the mean. For clarity, they are only depicted for the CATIE
 25 model. As shown by the figure, CATIE has the highest correlation with the empirical
 26 bias ($r=0.88$). These results are identical to those depicted in Fig. 3b. Full bias
 27 distribution for the observed bias and that predicted by the CATIE model, are
 28 presented in Fig. S2. Color scheme is identical to Fig. 1.



29

30 **Fig S4 – Distribution of model predictions.** Left, distribution of the 333,200 decision
 31 probabilities of participants' actions as predicted by the CATIE model and the 4
 32 versions of the QL model. Right, same for the logarithm of the probabilities.

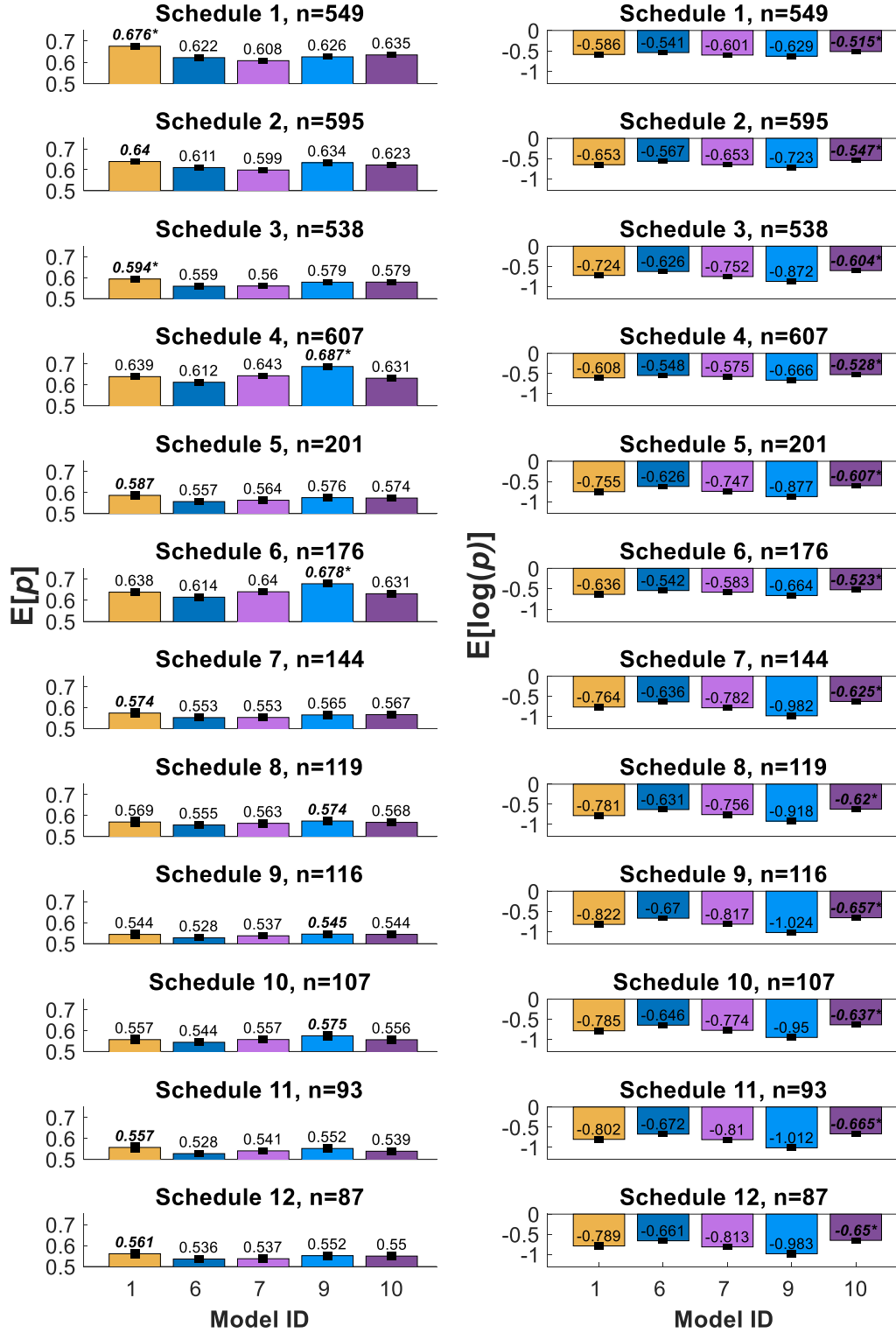


Fig S5 – Model prediction per schedule. Each bar depicts the average action probability (left) or log-probability (right), as predicted by CATIE (Model ID: 1) and four QL models (Model ID: 6, 7, 9, and 10). Color scheme is identical to that of Fig. 1. Each row summarizes the action probabilities of participants tested with a specific reward schedule ranging from schedule 0 (top) to schedule 11 (bottom). Error bars are standard errors of the mean, centered around the mean. The numbers are presented as text on top of each bar, with the maximal value per row highlighted in bold and italics. Asterisk denotes cases in which the maximal value is significantly larger than the runner up ($p < 0.01$, two-tailed t-test, uncorrected for multiple comparisons).

Predicted choice probabilities – between model comparison

Schedule ID	Difference of means	CATIE $E[p]$	$E[p]$	Effect Size (Cohen's d)	p-value	95% CI Lower	95% CI Upper
6	0.031	0.619	0.588	0.129	<0.001	0.03	0.032
7	0.027		0.592	0.095	<0.001	0.026	0.028
9	0.001		0.618	0.005	0.04	0	0.003
10	0.016		0.603	0.063	<0.001	0.015	0.017

Table S1. Comparison of predicted probability distributions between CATIE and other quantitative models (Companion to Fig. 4.a). The table contains the statistical details for the comparison of the predicted probability distribution of the schedule with the highest mean probability (CATIE) to those of other quantitative models. Two tailed p-value is computed using a permutation test with no correction for multiple comparisons. Confidence intervals are calculated using bootstrapping (See Methods).

Schedule ID	Difference of means	Schedule 10 $E[\log(p)]$	$E[\log(p)]$	Effect Size (Cohen's d)	p-value	95% CI Lower	95% CI Upper
1	0.109	-0.569	-0.678	0.186	<0.001	0.105	0.113
6	0.019		-0.588	0.05	<0.001	0.018	0.021
7	0.107		-0.676	0.21	<0.001	0.104	0.109
9	0.209		-0.778	0.296	<0.001	0.208	0.211

Table S2. Comparison of log predicted probability distributions between Schedule 10, and other quantitative models (Companion to Fig. 4.b). The table contains the statistical details for the comparison of the log predicted probability distributions of the schedule with the highest mean probability (Schedule 10) to those of other quantitative models. Two tailed p-value is computed using a permutation test with no correction for multiple comparisons. Confidence intervals are calculated using bootstrapping (See Methods).